AP PRECALCULUS

Unit 1 • Lesson 1.13 practice

Lesson 1.13 • Function Model Selection and Assumption Articulation

5 multiple-choice questions and 1 free-response question

Directions

Questions 1 to 5 are multiple choice: choose the best of the four answer choices. Question 6 is free response: show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

Data with equally spaced inputs have first differences $4.9,\ 5.1,\ 5.0,\ 4.9$. Which model is most appropriate?

(A) Linear
(B) Quadratic
(C) Cubic
(D) Rational

Question 2

NO CALCULATOR

Data with equally spaced inputs have second differences $-6.1,\ -5.9,\ -6.0$. Which model is most appropriate?

(A) Linear
(B) Quadratic
(C) Cubic
(D) Constant

Question 3

NO CALCULATOR

A company's average cost per item is $A(n) = \dfrac{500 + 4n}{n}$ dollars, for n items. What does $\lim_{n \to \infty} A(n)$ tell you?

(A) The fixed cost is \$500.
(B) The average cost becomes 0.
(C) The average cost per item approaches \$4.
(D) The cost per item increases without bound.

Question 4

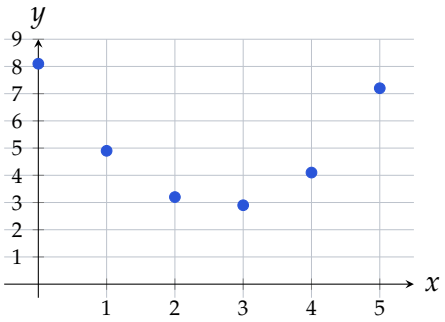
NO CALCULATOR

A quadratic model $P(t)$ gives the number of fish in a pond after t months; it has a maximum, then decreases. Which is the most reasonable restriction on the domain?

(A) All real numbers t
(B) $t < 0$
(C) Only integer values of P
(D) $t \geq 0$ and only while $P(t) \geq 0$

Question 5 (graph) NO CALCULATOR

A scatter plot of data is shown. Which type of function is the most appropriate model for the data?



(A) Linear
(B) Quadratic
(C) Constant
(D) Rational with a vertical asymptote

Question 6 (Free response) NO CALCULATOR

A phone plan costs \$20 per month, which includes 100 texts. Each additional text costs \$0.10.

(a) Write a piecewise function $C(n)$ for the monthly cost of sending n texts.
(b) Find $C(250)$ and interpret it.
(c) For $n > 100$, the average cost per text is $A(n) = \dfrac{C(n)}{n}$. Find $\lim\limits_{n\to\infty} A(n)$ and interpret it.